PURCHASE DESCRIPTION

# Liner, Cup - For Warhead, 2.75 Inch Rocket, M151 and XM229

## 1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the drawings, for one type of warhead explosive filler retainer, designated as Liner, Cup.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of' the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-A-2550 | Ammunition and Special Weapons, General Specification for |
| MIL-P-10025 | Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components |
| MIL-I-45607 | Inspection Equipment, Supply and Maintenance of |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-1168 | - Lot Numbering of Ammunition |
| MIL-STD-1235 | - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-171B(MR) | - Finishing of Metal and Wood Surfaces |

FSC:1340

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DRAWINGS

ORDNANCE CORPS

9220815 - Liner, Cup

PUBLICATIONS

ORDNANCE CORPS

ET-9220815 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring actitivy or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Cup Liner. - The cup liners shall comply with all requirements specified on Drawing (dwg.) 9220815 and with all requirements specified in applicable specifications.

3.3 Metal defects. - The components shall be free of surface defects, cracks, cracklike imperfections, pits, folds and other similar defects.

3.4 Workmanship. - The components and assemblies shall be fabricated and finished In a thorough, workmanlike manner. They shall be free of burrs, sharp edges, unblended radii, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent.

3.5 First Article Inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform



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any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product. - At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 50 cup liners which have been produced by the contractor and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production.

Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings/certificates of conformance for materials (see 4.1.1.e). A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspection to be performed. - Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the applicable drawings.



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4.2.3 Rejection. - If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements.

4.3 Inspection provisions for production

4.3.1 Lot formation. - The term “lot” as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing, not affecting safety, performance, interchangeability, or storage, as determined by the Government shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall not be less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period.

4.3.2 Examination. - Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL’s and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Once agreement has been reached as to which method will be used (group or individual) that method will be used for all of the CD’s listed in this specification, Equipment necessary for the performance of the inspections listed shall be In accordance with 4.3.4.



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4.3.2.1 Cup Liner (see dwg. 9220815).

| Categories | Defects | Method of Inspection | Code No. |
|---|---|---|---|
| Critical: | None defined. | | |
| Major: | AQL 0.65 percent | | |
| 101. | Outside diameter | Gage | 01001 |
| 102. | Angle | Gage | 01002 |
| 103. | Height | Gage | 01003 |
| Minor: | AQL 1.00 percent | | |
| 201. | Metal defective (see 3.3) | Visual | 01004 |
| 202. | Evidence of poor workmanship other than that covered under Minor 201 | Visual | 01005 |

4.3.4 Inspection Equipment. - Equipment Tabulation Number ET-9220815 Identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving Inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance. - Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment. - The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.



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## 5. PREPARATION FOR DELIVERY

5.1 Packaging and marking-Level C. - The cup liner shall be packaged and marked in accordance with Military Specification MIL-P-10025.

5.2 Liner cup packing material shall comply with PPP-B-636; Style RSC; grade 200 rein; carton size 15" x 15" x 11". Recommended quantity per carton, Bulk pack - 5000 Liner; fill voids with Kempak (or equipment).

## 6. NOTES

6.1 Ordering data. -Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers. -The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. - Inspection equipment lists Indicate the availability of inspection equipment designs by showing in the “number” column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as “Contractor Design” in the number column. These contractor designs must be approved by the Government prior to fabrication Or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.



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6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design Is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly, The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibilityty shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND9. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by



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the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army - MU

Preparing Activity:
Army - MU



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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